

Goals, Metrics, and Segmentation Hypotheses

1. Business goal

is recurring revenue

Why not the alternatives:

candidate metric	how it breaks
Users	has low impact if retention is not controlled
Activity	can raise with free trials/marketing, but doesn't necessarily affect ltv if retention is not controlled
Retention	goes up with no acquisition
Conversion	can raise with free trials/marketing, but doesn't necessarily affect ltv if retention is not controlled
Revenue per customer	increases if cheaper customers are lost

Revenue contains all drivers, improving one at another's expense shows up immediately

Guardrail

Every generation costs real compute, and video costs far more than an image → **revenue can be bought at a loss**

Goal: recurring revenue. Guardrail: margin per paying customer.

Revenue that arrives at negative margin doesn't count as growth

2. Drivers for revenue improvement

2.1 Definition

$Revenue_{this\ month} = \sum \text{over every past cohort} [cohort\ size \times share\ still\ active \times average\ payment]$

Why:

1. Revenue is not a flow. Bad month by acquisition doesn't affect instantly, only over time
2. Each cohort contributes from a different point on its own curve

2.2 How it moves month to month

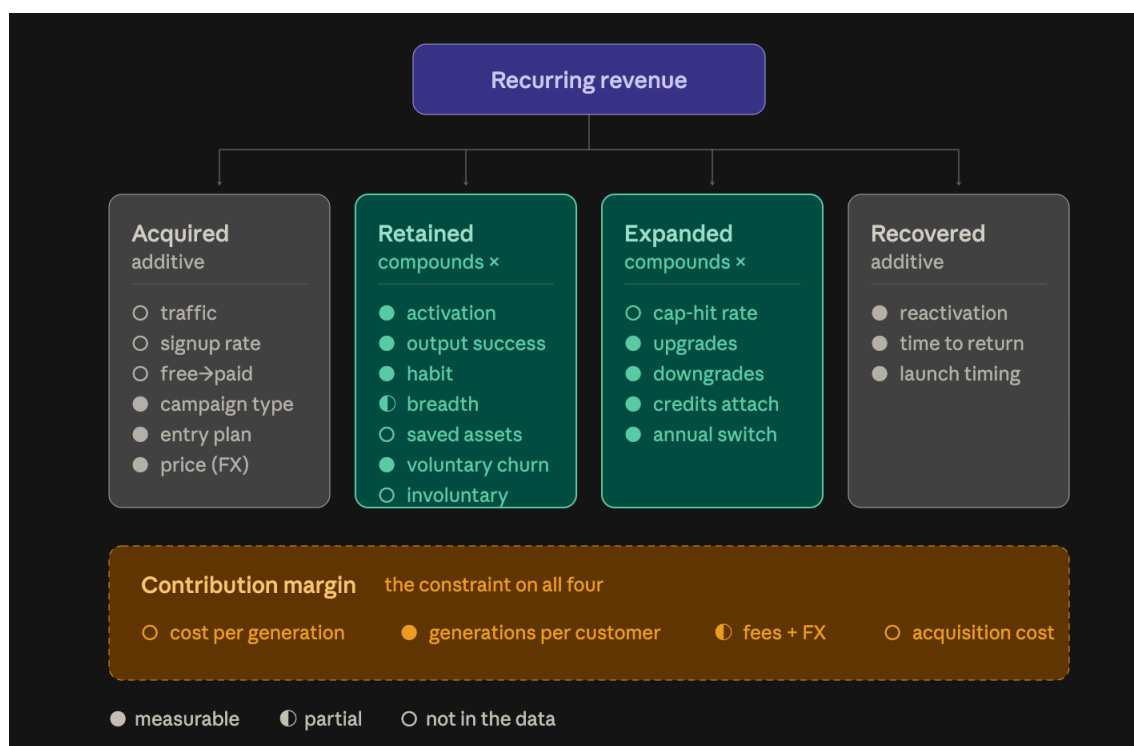
$Revenue_{\{month_t\}} = Revenue_{\{month_{t-1\}} \times Net\ revenue\ retention} + New\ revenue + Recovered\ revenue$

$Net\ revenue\ retention = (1 - churn) + (upgrades) - (downgrades)$

2.3 The four drivers

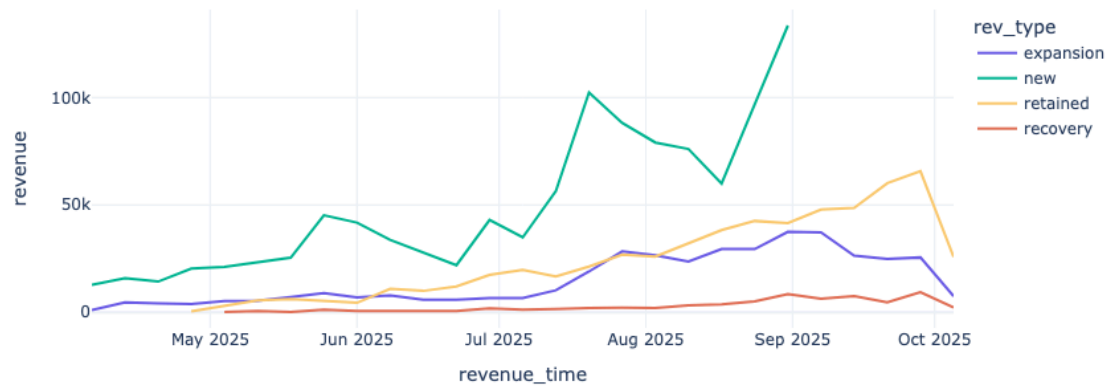
	driver	question	behaviour
1	Acquired revenue	How much new recurring revenue do we add?	additive
2	Retained revenue	How much of last month's revenue survives?	compounds — multiplies every month
3	Expanded revenue	How much more do survivors pay over time?	compounds, on the survivors
4	Recovered revenue	How much comes back from customers who left?	additive, small
-	Contribution margin	What does it cost to serve any of this?	a constraint

4. The metrics tree

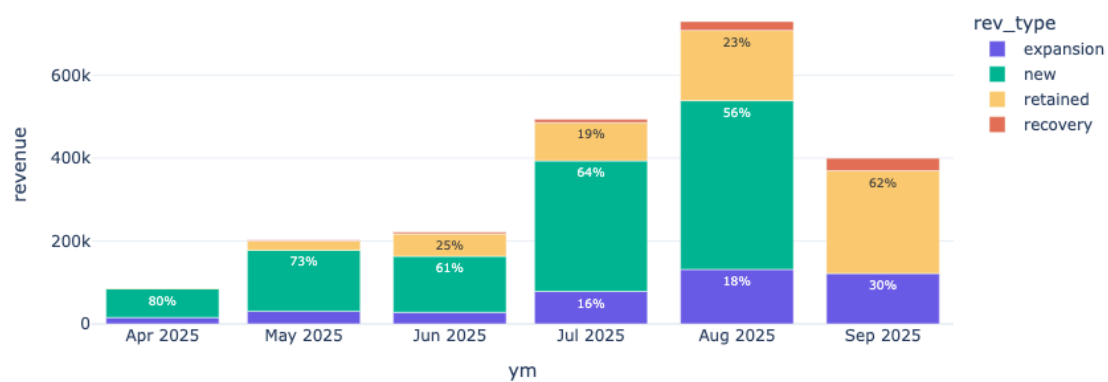


Analysis based on the data

weekly revenue by driver



monthly revenue by driver



Conclusions

- New revenue dominating means the business is running on acquisition
- Rise in retention revenue could mean product development
- Expansion revenue sets a ceiling on how much an upsell programme could be worth

Prioritization

Considering

- the case context (retention study and LTV growth)
- constraints (all users in the data are subscribed, so we cannot make any conclusions based on the top of the funnel "creation of account -> subscription")
- share of each type of revenue (overall acquisition 50%, retention 28%, expansion 19% and recovery 3%)
- unavailability of the constraint of the optimization problem (CAC, cost per generation, fees)
- retention and expansion being multiplied by the whole base of users regardless of acquisition

case context	retention + LTV, not acquisition campaigns
funnel	everyone in the data already pays — no signup → subscribe
mix	new 50% / retained 28% / expansion 19% / recovery 3%
costs	CAC, compute per gen, fees are not in the data
math	retained and expansion \$ multiply the whole base, every month

Focus: (1) retention (2) expansion. Acquisition is the largest slice but we cannot see the unpaid funnel, and recovery is too small.

6. Retention and expansion analysis

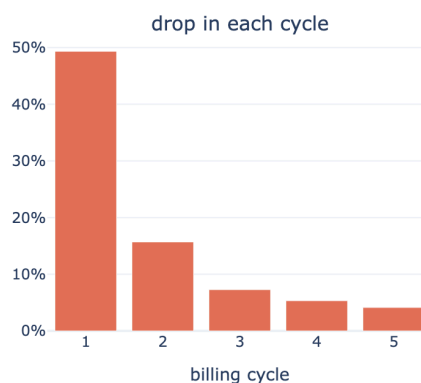
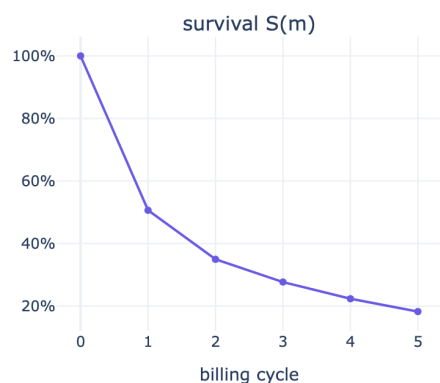
Retention

How renewal is measured (revised in `LTV Modeling.md` ; applies to every renewal number in this file)

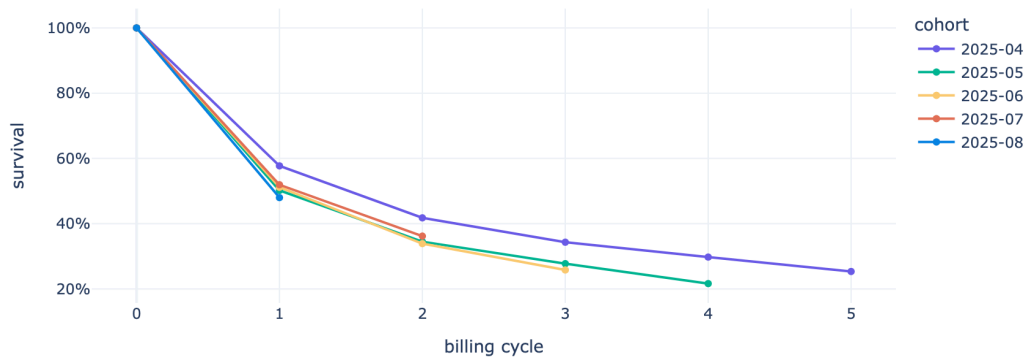
rule	value
billing cycle	30 days from the first charge, grid shifted by a 2-day tolerance — no grace period
renewal	a non-Reactivation subscription charge inside the cycle (win-backs are recovery, not retention)
observability	the cycle must have fully elapsed → tenure ≥ 32d
annual	excluded from renewal rates (87% never pay twice, so a monthly renewal is undefined for them)

- Target retention period is 1 month

where the curve bends

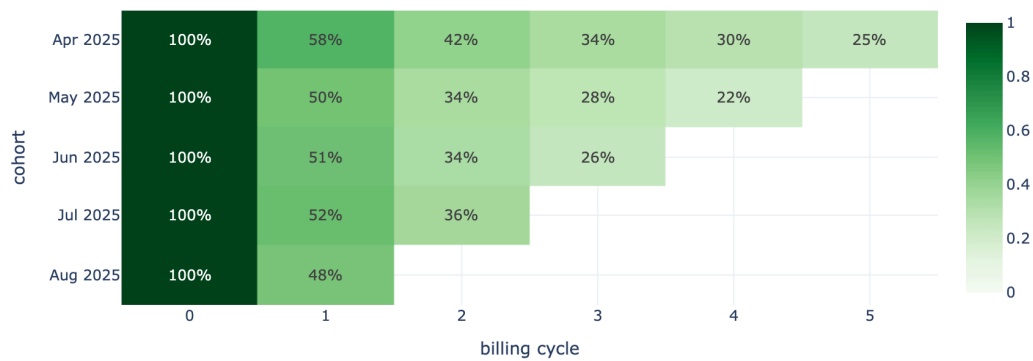


retention curves by cohort



- Retention by cohorts

paid retention by cohort, billing cycles since first charge



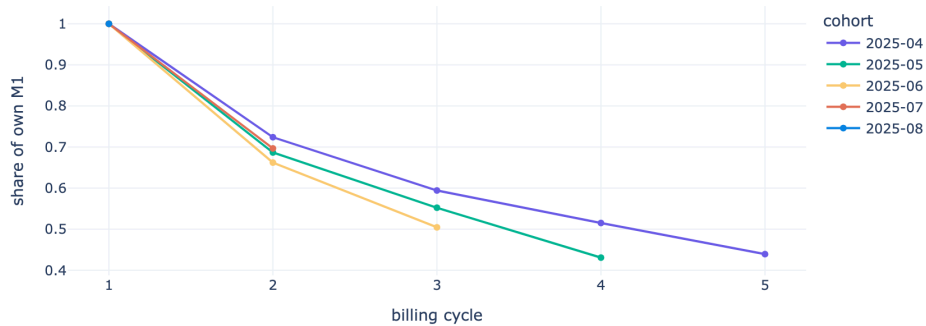
Where the churn is? The first cycle drops **49.4%** of monthly customers — more than every later cycle combined — so a 30-day window captures nearly all the signal and longer windows mostly add variance.

the first renewal (30 days) is the target period. It carries the highest churn, sample size and is early to act on. Effectively most of the retention uplift is concentrated in the first 30 days

- how to model features for retention: pool all 1-month users together or separately and why?
 - is the difference between cohorts a level shift or a shape change? (level shift -> same product, different customer quality, shape change -> the product itself changed users)

behavior)

survival rescaled to each cohort M1 - do the shapes overlap?



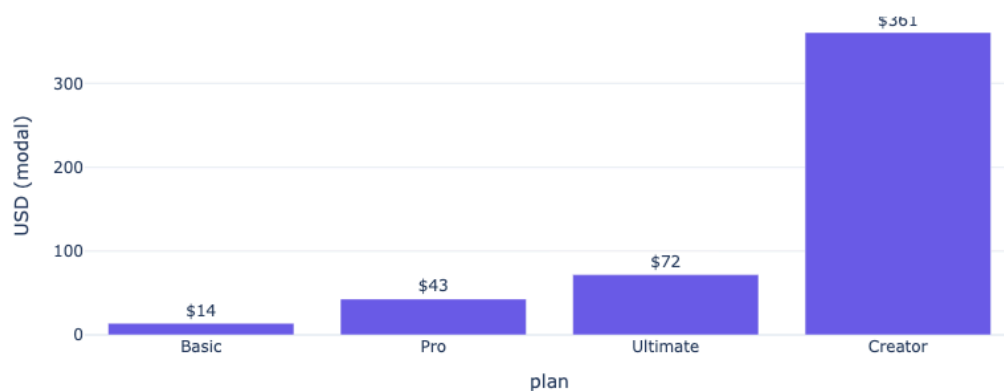
Cohorts differ on level: first renewal is 58% (Apr), 50% (May), 51% (Jun), 52% (Jul), 48% (Aug) — a 10 pp spread, with the newer cohorts at the bottom. But once each cohort is rescaled to its own M1, the curves sit close together: M2 lands at 0.66–0.72 of M1 and M3 at 0.51–0.59 — 6–9 pp of spread, against 17 pp under the loose 37-day window. Apr-25 is still the best cohort at every horizon (M1 58%, M4 30%), and it decays only slightly slower.

The difference between cohorts is customer quality, not how fast survivors decay. Since it is a level shift, cohorts can be pooled for first-renewal modelling with the cohort itself as a feature to absorb the level. What must stay separate is **first-renewal (M1)** vs **post-M1 decay**: their drivers disagree (rank corr 0.28, §9), so the same early behaviour still maps to different later outcomes

Expansion

We consider the plan logic as Basic < Pro < Ultimate < Creator based on pricing difference, consistent over regions.

plan ladder by modal monthly create price

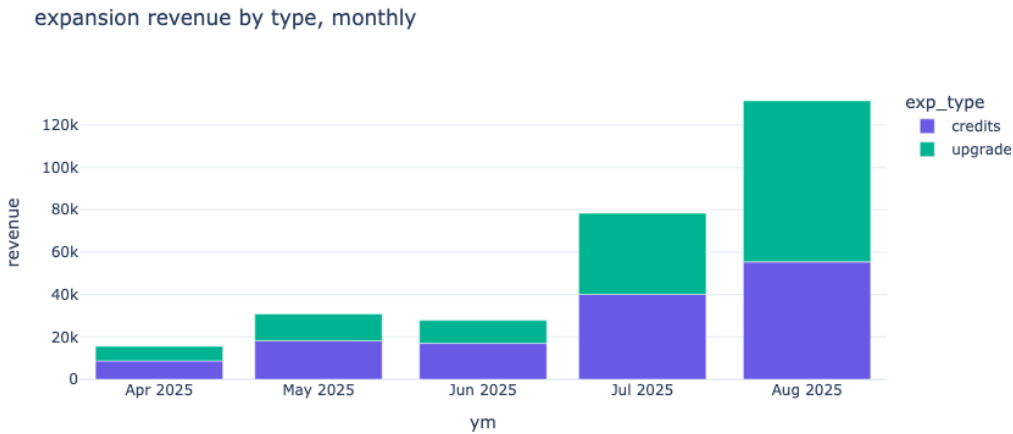


Let's define expansion as

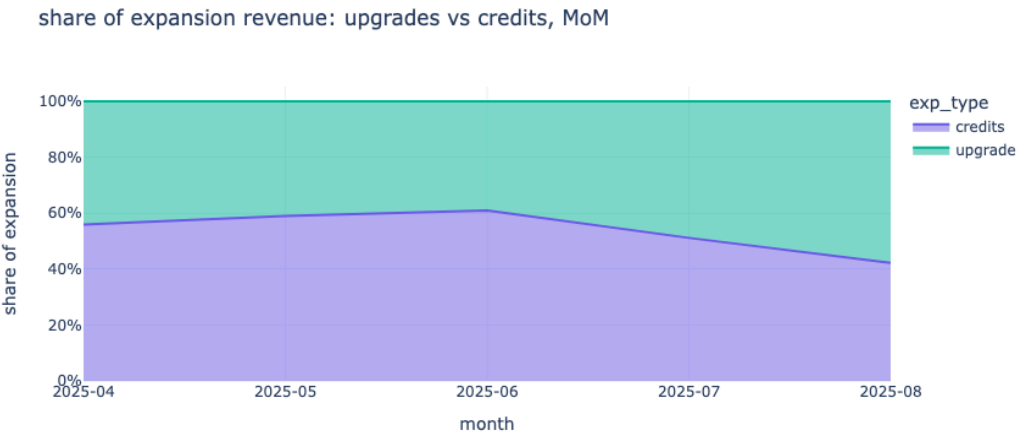
#	type	definition	revenue	count
1	Subscription upgrade	rise in subscription plan (where 'first_plan' is the original plan paid for by user)	210,775.88	3306
2	Credits top-up	Considered here as non-dependent on any subscription plan, therefore is another part of expansion	194,794.23	6773

Share of each type of expansion month over month

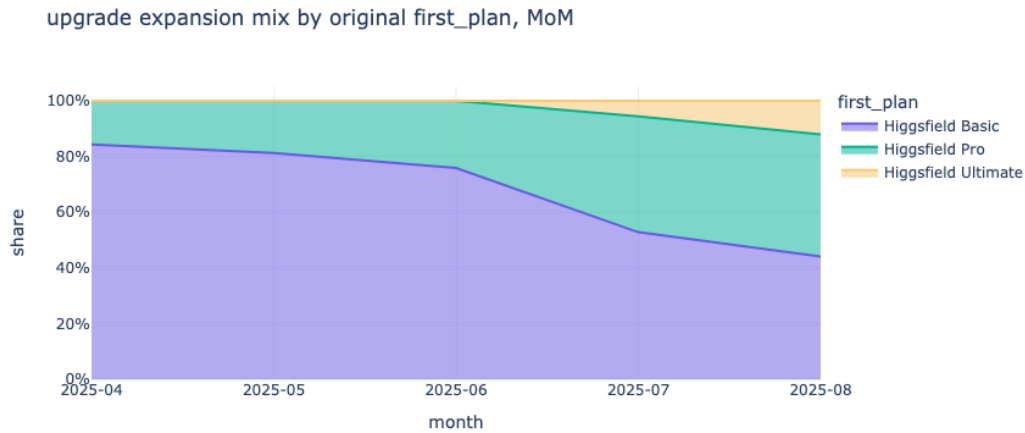
- absolute



- relative by types



- expansion is different by type of subscription plan



- when does expansion happen in the user's journey with the product?

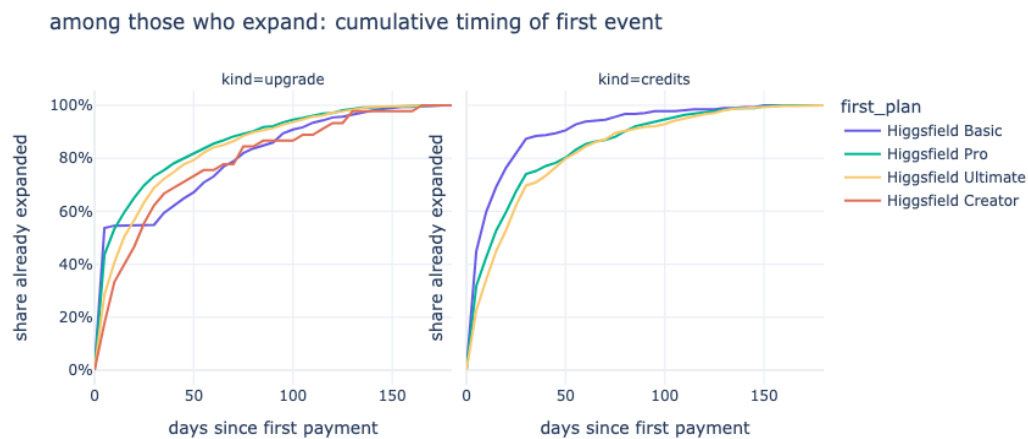


1. Expansion is early — but it differs by entry plan. Medians are all inside the first ~3 weeks. Basic upgrades are immediate (median 0d); Pro ~8d; Ultimate ~15d; Creator ~21d. Credits lag upgrades slightly on every plan except Creator. So the modelling window should be plan-specific
2. Basic is a different product story. Same-day upgrades + fast credits → many Basic buyers are under wrong plan at signup. Mean upgrade delay (33d) >> median (0) → a fat right tail: immediate upgraders vs people who wait ~a billing cycle. Basic should be split into day-0 upgrades vs later — we need to drive 0-day conversion from Basic to upgrade
3. Volume and timing are not the same ranking. Pro/Ultimate dominate upgrade counts; Basic dominates speed. Creator barely expands via upgrade (top of ladder) and has tiny credits n=18 — treat Creator as credits-only and don't pool it with Basic/Pro for upgrade models.

Days to **first** expansion event (by type × original plan):

first_plan	kind	n	median days	mean days	focus
Basic	upgrade	350	0.0	33.4	Primary — same-day misfit; split day-0 vs late tail
Basic	credits	276	6.1	15.8	Early cap pressure after signup
Pro	upgrade	1,404	7.9	24.1	Primary volume — week-1 upgrade window
Pro	credits	581	13.9	26.5	Week 2–3 credits; separate from upgrade model
Ultimate	upgrade	1,019	14.9	28.4	Primary volume — ~2-week window, not day-0
Ultimate	credits	482	18.1	29.9	Similar to Pro credits
Creator	upgrade	45	20.9	36.9	Ignore for upgrade models (top of ladder, tiny n)
Creator	credits	18	17.5	23.4	Credits-only route; n too small to model alone

- what drives credits top up and upgrade of subscription - how are user journey different in these cases?



Credits and upgrades are different journeys. Credits medians are later and the Pro/Ultimate credits bases are large. We shouldn't collapse into one "expanded" label for feature work — separate targets: first upgrade vs first credits (and "both")

6. CJM

Paid journey. Starts at first charge.

PAY → FIRST GEN → WEEK-1 HABIT → FIRST-MONTH HABIT → upgrade / credits → FIRST RENEWAL → STAY

↳ churn

#	stage	when	what happens
1	Pay	day 0	Chooses plan and pays
2	First generation	hours–day 1	Runs first job
3	Week-1 habit	days 1–7	Comes back on more than one day
4	First-month habit	days 8–30	Keeps using until renewal (gaps stay short)
5	Expansion	mostly days 0–30	Upgrades plan and/or buys credits (optional; overlaps 3–4)
6	First renewal	~day 30	Pays or churns
7	Stay	month 2+	Keeps renewing; may expand further

Why week 1 **and** month 1 (not only one of them):

window	role	why it is in the CJM
Week 1	early signal	Fast to measure, high impact; see hypothesis check
Week 2–4 (month 1)	path to renewal	First renewal is ~day 30; <i>days_28</i> / gaps over the full month predict renewal better than week 1 alone; see hypothesis check

7. Segmentation hypotheses

Category	Hypothesis	Product Hypothesis (H1)
Retention	Activation Velocity	Faster time-to-first-value (generating an output within 1 day of sign-in) significantly increases Month 2 retention.
Retention	Early Stickiness	Building a multi-day habit early (active ≥3 days in Week 1) prevents early churn and drives significantly higher 1-month retention.
Retention	Day-of-Week Cohort	Monday/Tuesday sign-ups retain better in Month 1 than weekend sign-ups, indicating that work-week intent drives stickier usage.
Retention	First-Day Binge vs. Pacing	Spaced usage is stickier than a single exploratory binge; users concentrating >80% of Week 1 volume on Day 1 have significantly lower retention.
Ret & Exp	Professional Use Case	A high weekday-to-weekend generation ratio strongly predicts professional utility, driving higher volumes and willingness to pay (expansion revenue).

Ret & Exp	Working Hours vs. Evening	Generation activity concentrated within standard working hours (9 AM - 5 PM) is a stronger predictor of subscription tier upgrades than evening usage.
Ret & Exp	Routine Consistency	Integrating the tool into a predictable daily routine (low variance in session start time) significantly reduces 1-month churn.
Ret & Exp	Activity Cool-Off	A week-long lapse (activity gap of ≥ 7 consecutive days) in Month 1 is a critical warning signal that predicts Month 2 churn.
Ret & Exp	MoM Growth	Users organically expanding their usage ($>10\%$ growth Month-over-Month in the first 60 days) are significantly more likely to trigger credits top-ups.
Ret & Exp	End-of-Cycle Spike	A volume spike in the final 3 days of a billing cycle predicts a following-month top-up, as users suppress usage while waiting for free renewal.
Ret & Exp	Volume-Driven Expansion	Hitting early usage ceilings is the primary trigger for near-term expansion; top 20% Month 1 volume predicts Month 2 top-ups/upgrades.
Ret & Exp	Platform Superiority	Desktop facilitates deeper workflows, resulting in a significantly higher retention rate than Mobile-only usage.
Ret & Exp	Platform-Expansion	Desktop users generate higher average expansion revenue over 6 months, as the platform is more conducive to paid professional work.
Ret & Exp	Annual vs. Monthly	Upfront commitment correlates with deeper adoption; annual subscribers maintain a significantly higher baseline of weekly generations by Month 2.
Ret & Exp	Time to First Expansion	Breaking the payment friction early creates repeat buyers; expanding within the first 30 days strongly predicts future top-ups.
Ret & Exp	Regional Tier Adoption	Localization impacts lifetime value ceilings; specific geographic regions yield a significantly higher proportion of premium tiers and top-ups.
Expansion	Prompt Evolution	Evolving beyond basic prompts (positive growth in prompt length from Week 1 to Week 4) indicates usage skill and readiness for upgrade.
Expansion	Frustration/Low-Effort	Persistent short prompts ($>30\%$ under a minimum word count in Month 1) signal a failure to find value, leading to significantly lower LTV.
Expansion	Complexity Plateau	Finding a reliable, repeatable use case (prompt length variance dropping and stabilizing in Months 2-3) locks users into long-term retention.
Expansion	Volume vs. Length	The most valuable power users combine high-frequency usage with high-effort inputs, generating significantly higher LTV than

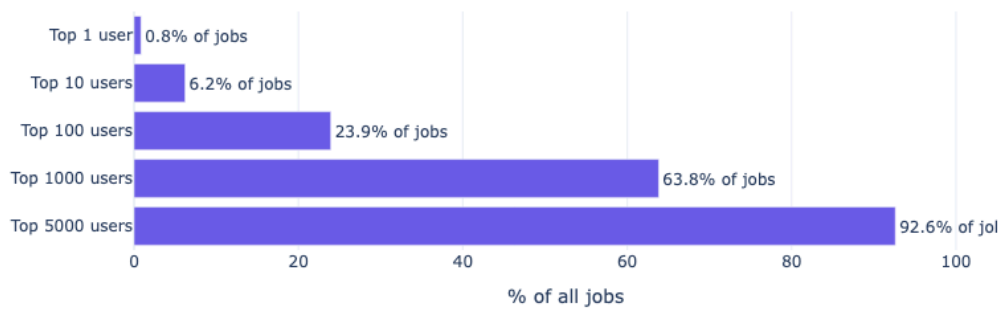
high-volume/short-prompt users.

8. Constraints & Guardrails

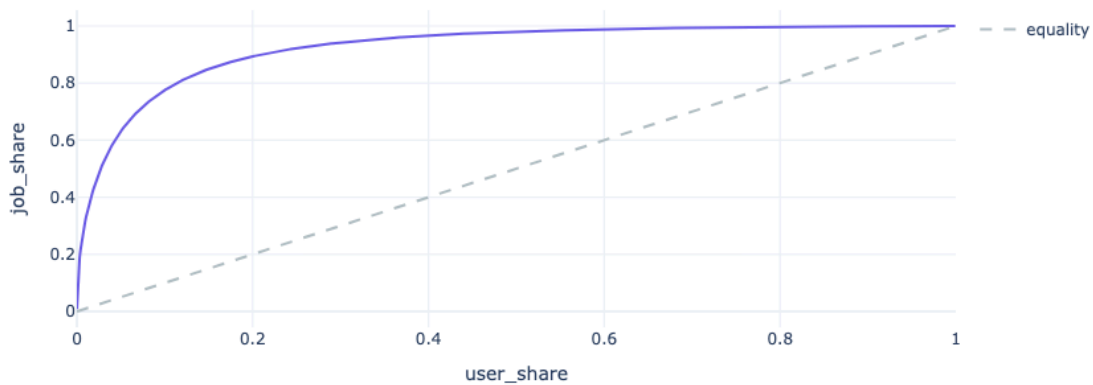
While the hypotheses above map the path to higher retention and expansion, they must be evaluated against the fundamental unit economics of Generative AI. In this business model, every generation incurs a hard compute cost, meaning revenue growth can easily be bought at a negative margin.

The Power-Law Usage Distribution Based on the cumulative job share data, the distribution of usage is extremely left-skewed:

Cumulative job share: top N users together



job concentration (lorenz)



This reveals that the bottom ~74% of users consume only 7.4% of the system's total jobs. Because exact CAC, server costs per generation, and API fees are unobservable in the current data, we must operate under constraints:

- The "Commercial Gym" Subsidization Model: The bottom 74% of low-volume users are highly profitable and effectively subsidize the extreme costs of the top 5%. If product interventions

successfully push the bottom 74% to adopt the high-volume habits of the top 5% without moving them to usage-based billing, gross margins will collapse.

- Negative-Margin Power Users: Under flat-rate subscription tiers (Basic/Pro/Ultimate) without hard caps, the top 5.2% of users driving 64% of the volume are likely operating at a negative net margin. High activity only equates to high value if it triggers an expansion event.
- The goal of retention interventions for the broad base (the 74%) should be to build low-volume, high-value stickiness (e.g., getting them to return once a week for a specific task), rather than purely maximizing their daily generation count.
- Expansion (upgrades and credits) must be targeted at the top 25% of users. For this cohort, expansion is not just an LTV growth lever; it is a mechanism required to offset their extreme compute burn and force them back into profitability

9. Hypothesis testing

Statistical testing

Pooled target

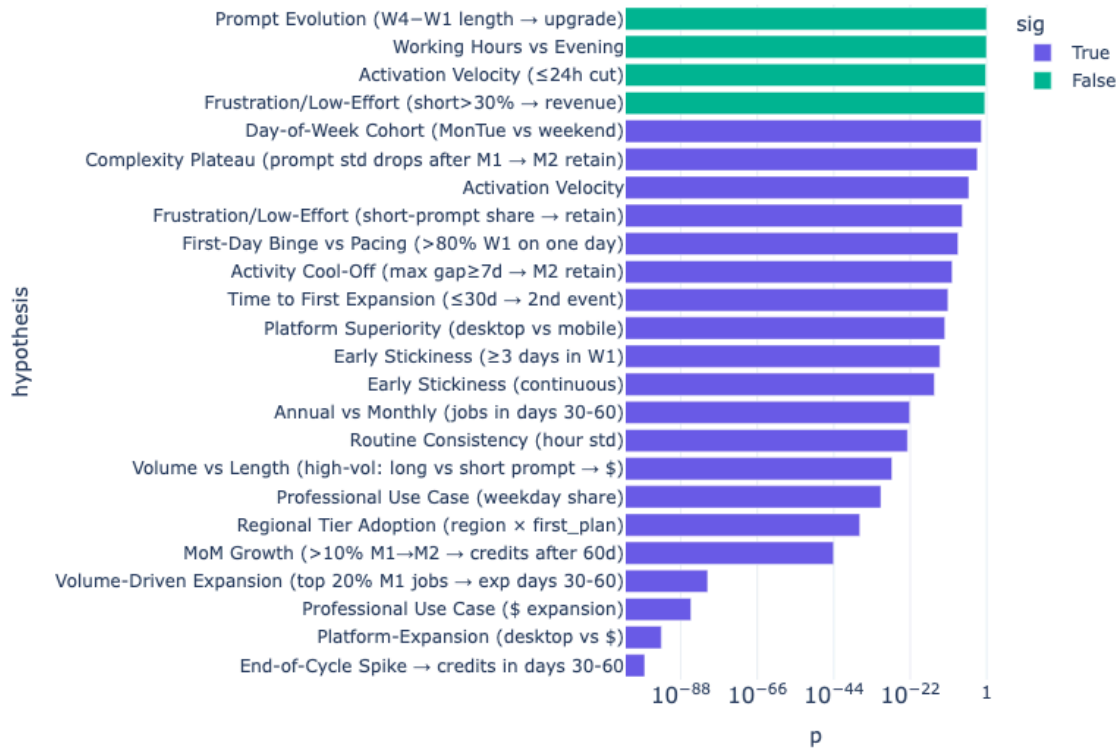
One feature vs one outcome. Sample = monthly subscribers who have been around long enough to observe the outcome. Features for 30-day outcomes use only generations in the first 28 days after first charge.

Cool-off / MoM / second expansion use a later window and a later outcome.

	test	effect
continuous	Mann–Whitney U	rank-biserial, -1 to +1
binary / category	chi-2	Cramér's V, 0 to 1

supported = $p < 0.05$ and the sign matches H1. opposite = significant, wrong sign. n is large, so $p \approx 0$ is cheap — rank by |effect|.

hypothesis tests: p-values (log scale)



hypothesis	verdict	effect
End-of-cycle spike → credits (d30-60)	supported	0.29
Volume vs length (high-vol, long prompt → \$)	opposite	-0.21
MoM growth → credits after 60d	supported	0.19
Top 20% M1 jobs → expansion d30-60	supported	0.18
First expansion ≤ 30 d → second event	supported	0.16
Annual vs monthly (jobs d30-60)	supported	0.15
Desktop → expansion \$	supported	0.14
Weekday share → expansion	opposite	-0.13
Weekday-high → expansion \$	opposite	-0.13
Hour variance → retention	opposite	0.10
Region × plan mix	supported	0.08
Early stickiness (days in W1, continuous)	supported	0.07

Prompt std drops after M1 → M2 retain	supported	0.07
Early stickiness (≥3 days W1)	supported	0.06
Cool-off (gap ≥7d → M2 retain)	supported	0.06
Desktop vs mobile retention	supported	0.06
Binge >80% of W1 on one day	supported	0.05
Short-prompt share → retain	opposite	0.04
Hours to first gen → retain	supported	−0.04
Mon/Tue vs weekend signup	supported	0.02
Short prompts >30% → revenue	not supported	0.02
Generated within 24h	not supported	0.01
Working hours vs evening → upgrade	not supported	0.00
Prompt evolution W4–W1 → upgrade	not supported	0.00

What to keep

- retention: active days in week 1, gaps, not bingeing. Time to first gen is real but small; the 24h cut is not.
- expansion: month-1 volume, end-of-cycle spike, MoM growth, expand early. Desktop and plan/region mix.

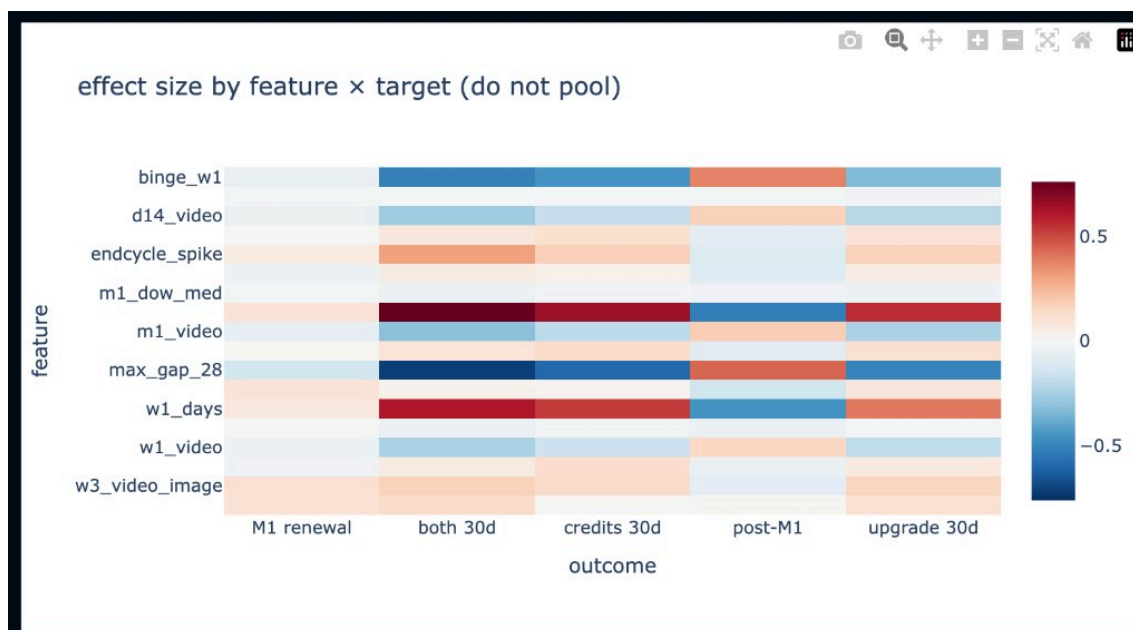
What to drop

- 9–5 / weekday as “professional” (weekday-heavy expands less)
- prompt length as skill / LTV (evolution ns; long prompts on high volume earn less)
- hour-of-day routine
- 24h activation rule

Possible cuts for segments: week-1 active days, month-1 volume and gaps, desktop vs mobile.

Target segmentation: five labels

	n	base
M1 renewal	16,091	50.9%
post-M1 (already renewed at 30)	5,610	97.3%
upgrade 30d (can still move up)	15,116	10.6%
credits 30d	16,091	9.8%
both 30d	16,091	2.4%



feature	M1	post-M1	upgrade	credits	both
m1_jobs	0.10	-0.53	0.56	0.65	0.76
w1_days	0.07	-0.45	0.40	0.53	0.61
max_gap_28	-0.14	0.44	-0.51	-0.60	-0.71
binge_w1	-0.06	0.38	-0.33	-0.45	-0.52
endcycle_spike	0.06	-0.09	0.17	0.18	0.31
m1_video	-0.06	0.19	-0.24	-0.20	-0.31
m1_video_image	ns	ns	0.12	0.13	0.09
wow_growth	0.11	ns	0.11	ns	0.13
*_dow_med	ns	ns	~0	ns	ns

What actually differs

- M1 vs expansion is size, not sign. Habit and volume point the same way as in the table above; expansion effects are 4–6× larger (jobs 0.10 vs 0.56–0.76). First renewal is a weak-signal outcome; expansion is a volume outcome.
- post-M1 flips the signs. Among people who already paid a second time, the 152 who then miss month 2 were the *heavy* M1 users (median 76 jobs vs 17). That is a different group from M1 churn — not a reason to reverse the M1 keep-list
- upgrade / credits / both are one axis, not three journeys. Same signs, intensity both > credits > upgrade (jobs 227 / 104 / 79). "Both" is the extreme of the same habit+volume cut, 2.4% of the base.
- Video share vs video/image ratio are not the same feature. All-video users (share ≈ 1, no image jobs) expand *less*. Among people who mix, a higher video/image ratio expands *more*. Median weekday of

jobs does nothing.

Do we need this split?

- Retention: yes, do not pool M1 with post-M1 — the signs disagree, so one “retained” label would cancel. Impact-vice, we stay on M1. post-M1 is not a second equally-weighted target
- Expansion: no extra feature ranking from splitting upgrade vs credits. Direction matches the pooled expansion tests above

Feature importance using catboost and shallow trees

Here we decided to not only test hypotheses, but to do complete feasible feature generation and use ML to identify possible cuts for segments

Features description

All generation features below are from days 0–28 after first charge, unless noted. Categorical: platform, region, first_plan.

ML: pooled models sit next to the split ones (M1 renewal; any expansion 30d). Split labels are post-M1, upgrade, credits, both.

target	sample	y
pooled: M1 renewal	monthly, tenure $\geq$ 30d+grace	renewed_30
post-M1	already renewed at 30, tenure $\geq$ 60d+grace	renewed_60
pooled: any expansion	monthly, tenure $\geq$ 30d+grace	expanded_30
upgrade	monthly, can_upgrade, tenure $\geq$ 30d+grace	upgraded_30
credits	monthly, tenure $\geq$ 30d+grace	credits_30
both	upgrade and credits in 30d (skipped if thin)	both_30

group	feature	what it is
habit	hours_to_first_gen / days_to_first_gen	time from first pay to first job
	never_generated	no jobs in 28d
	w1_days, d14_days, m1_days	distinct active days in 7 / 14 / 28d
	w1_jobs, d14_jobs, m1_jobs	job counts in the same windows
	binge_w1	share of week-1 jobs on the single heaviest day
	jobs_per_day_var / mean	spread vs level of jobs per active day
	weekday_weekend_ratio, m1_weekday	weekday vs weekend usage
	m1_workhour	share of jobs in 9–17 (UTC, not local)
	m1_hour_std	how peaked the hour-of-day is

	m1_dow_med, w1_dow_med, d14_dow_med	median weekday of jobs (0=Mon ... 6=Sun, UTC)
modality	m1_video, w1_video, d14_video, w3_video	video share of jobs by window
	m1_video_image, w1_video_image, d14_video_image, w3_video_image	video jobs / image jobs in the same windows
	video_image_wow	w3 video/image – w1 video/image
prompt	m1_prompt_med / max	typical and longest prompt in 28d (chars, not words)
	m1_short	share of prompts $\leq$ p10 of non-empty prompt_length (set from the 28d distribution)
	prompt_evolution	mean prompt length week 3 – week 1
momentum	wow_growth	(jobs week 3 – week 1) / week 1
	cooloff_last10	zero-job days in the last 10 days of month 1
	recency_at_28	days from last job to day 28
	max_gap_28	longest silent stretch in month 1
	endcycle_spike	jobs in last 3 days vs daily mean
plan / \$	quota_burn_d15	jobs by day 15 $\div$ p75 of 28d jobs on the same plan (no real quota in the data)
	frontload_d15	jobs by day 15 / jobs in 28d
	annual	first period is year
	days_signup_to_pay	created_at $\rightarrow$ first charge
	rungs_above, can_upgrade	steps left on Basic < Pro < Ultimate < Creator
context	desktop	customer.platform = desktop (one platform per account)
	signup_dow, signup_month	day-of-week and month of created_at
quality	m1_fail	fail rate in 28d

Why SHAP and permutation

	SHAP (mean SHAP)	permutation (Δ AUC)
question	how much does this feature move the score when everything else is in the model	does randomizing it on the test set hurt ranking

good for	seeing what the tree actually splits on, including correlated usage features	deciding what to drop
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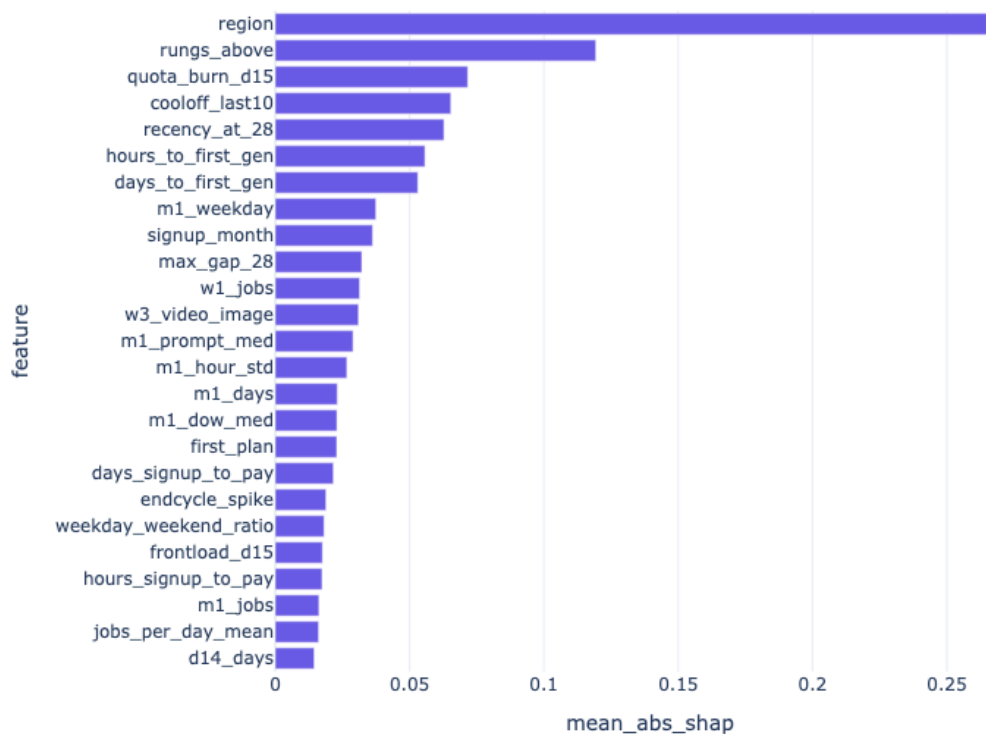
Permutation CI: if the 95% interval includes 0, the drop is not distinguishable from noise. Treat as not significant even if the point estimate is slightly positive. Most of the top-20 perm lists are in that bucket.

We trust a feature if the bivariate sign is right **and** perm CI excludes, and use SHAP to see the rest of the correlated bundle (jobs / days / cool-off / recency), not as a second significance test.

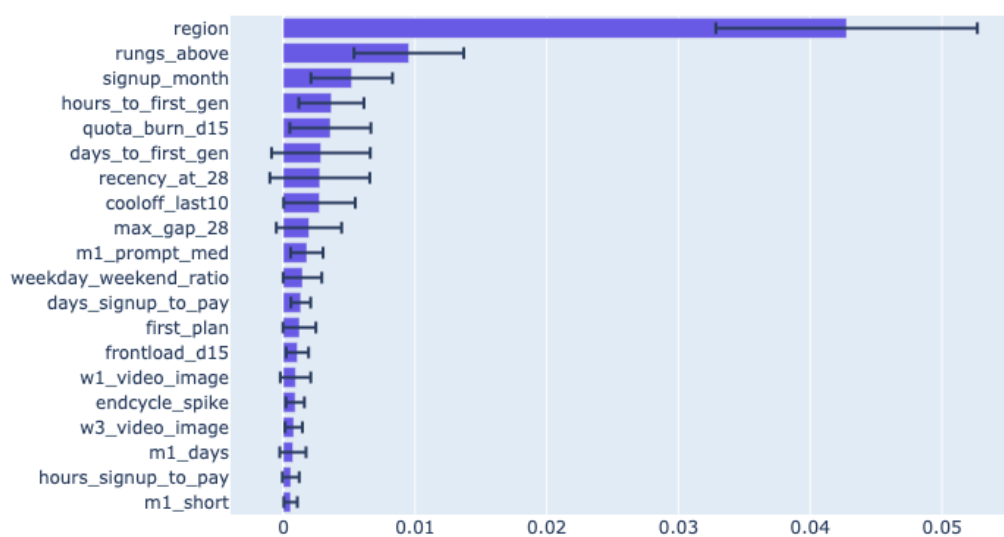
Features to use

1. Retention — first renewal (M1) model

CatBoost SHAP — pooled: M1 renewal



permutation Δ AUC — pooled: M1 renewal



We keep

- mix: `region` , `rungs_above` , `signup_month` — only features with a clear perm drop
- activation: `hours_to_first_gen` (test -0.04)
- volume vs plan: `quota_burn_d15` . `m1_jobs` is real in MW (0.10) but redundant with plan

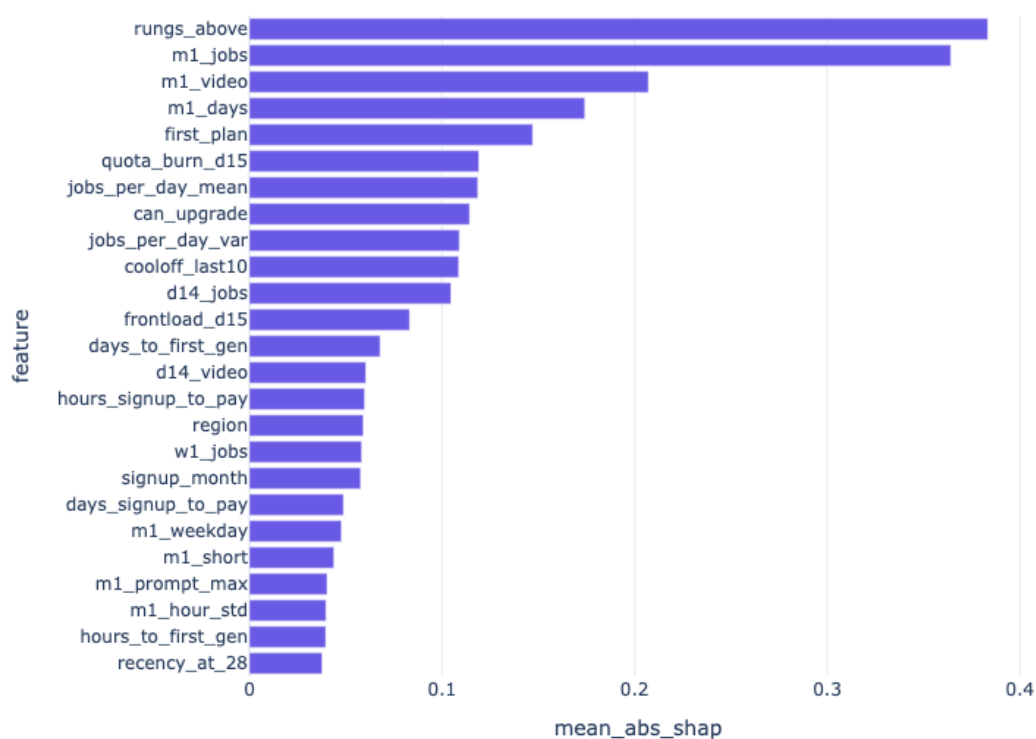
Segment on (tests, not unique AUC): week-1 days (0.07), gaps (-0.14), not bingeing (0.05). SHAP shows them as one bundle (`cooloff` , `recency` , `max_gap` , `w1_days`); perm CI includes 0 on each alone.

Drop: prompt length (`m1_prompt_med` is perm-sig, tests opposite 0.04 / -0.21), weekday / hour / `*_dow_med` (hour variance opposite 0.10), post-M1 as a second model.

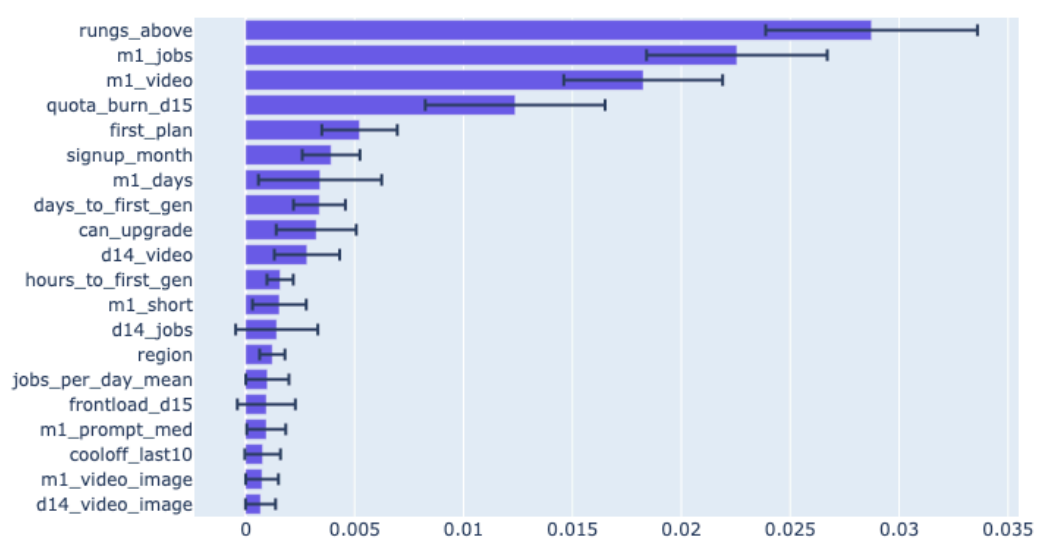
First renewal is mix + a small habit signal, not volume. Approximately same results as in stat tests

2. Expansion — pooled model is enough for measurement

CatBoost SHAP — pooled: any expansion 30d



permutation Δ AUC — pooled: any expansion 30d



Keep

- subscription plan: `rungs_above` , `first_plan` , `can_upgrade` — largest unique perm drop
- volume: `m1_jobs` (0.02 [0.02, 0.03]), `quota_burn_d15` (0.01 [0.01, 0.02]). Tests already: jobs 0.56–0.76, top 20% M1 jobs → later expansion 0.18, end-cycle spike 0.17–0.31.
- modality: `m1_video` (0.02 [0.01, 0.02]) — all-video expands /less (tests –0.20 to –0.31). Share survives perm; video/image ratio does not.

Pitch (rank corr upgrade vs credits 0.53, not 0.73): upgrade = where you sit on the ladder; credits = how hard you used the plan (`m1_jobs` , `quota_burn` , video share). “Both” is the 2.4% tail of the same volume cut, not its own feature set.

Drop: `*_dow_med` , prompt evolution (ns), 9–5 / weekday-as-professional (tests opposite –0.13), `d14_jobs` and `frontload_d15` (perm CI includes 0). SHAP’s extra volume rows (`m1_days` , `jobs_per_day_*` , `cooloff`) are the same axis as `m1_jobs` — interchangeable, not extra cuts.

Expansion is volume + ladder. Tests and ML agree. Do not build three expansion models for ranking.

Is the split needed? (ML)

	rank corr	perm
M1 vs post-M1	0.28	different. post-M1 $\approx$ <code>rungs_above</code> only
pooled exp vs upgrade	0.73	same volume+ladder core
pooled exp vs credits	0.73	same
upgrade vs credits	0.53	ladder vs usage intensity
M1 vs pooled exp	0.41	overlap on plan; expansion needs volume, M1 needs region

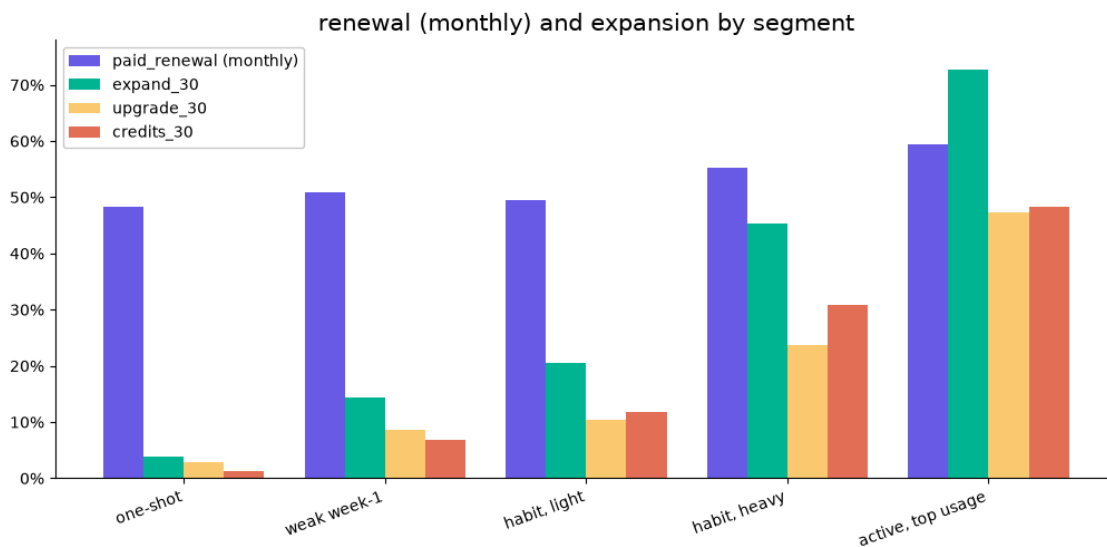
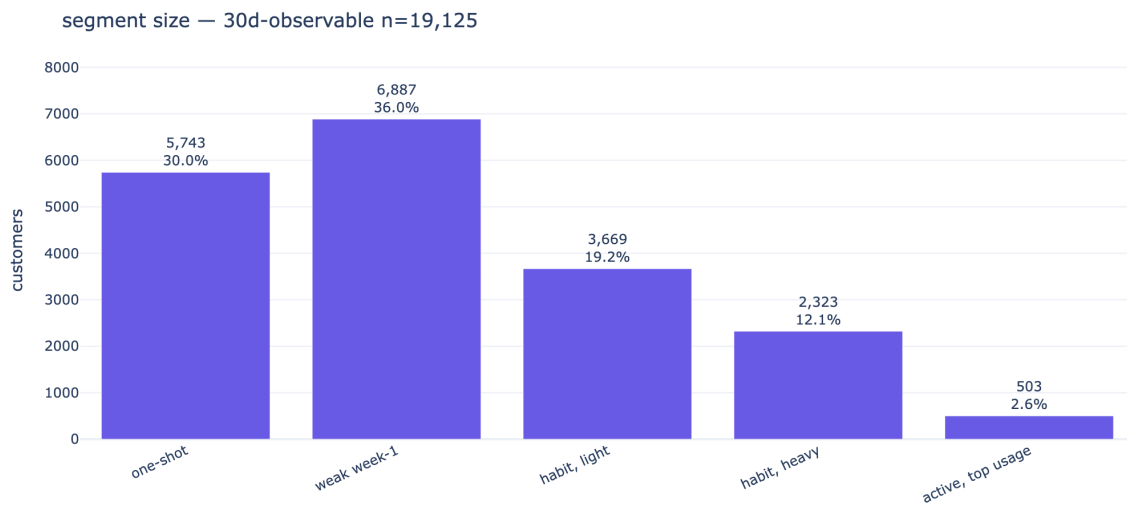
Same call as the tests

Retention: we should not pool M1 with post-M1. Expansion: no need for three models for feature ranking.

10. Segmentation

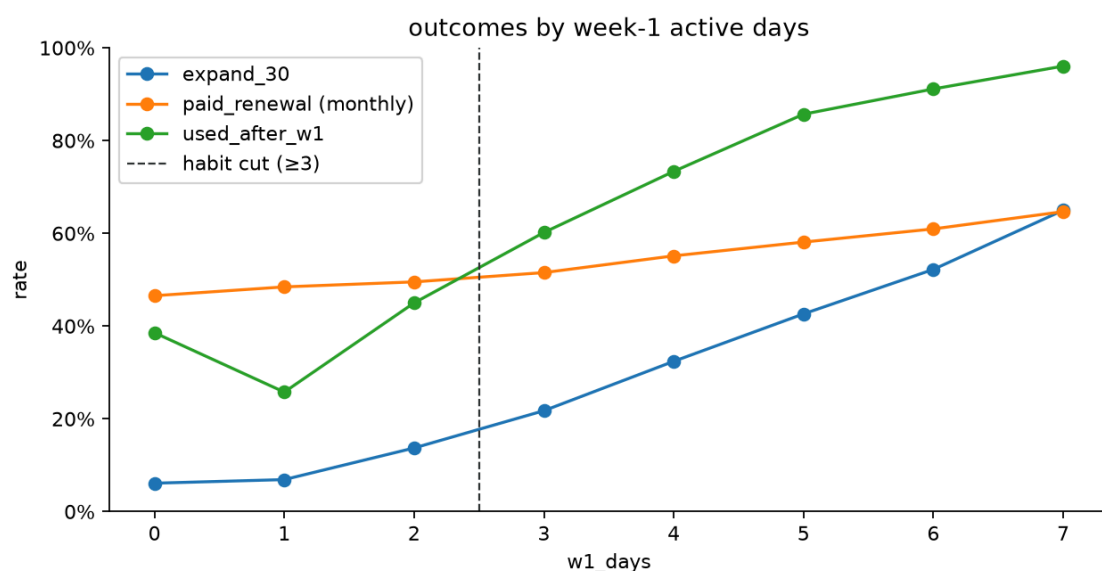
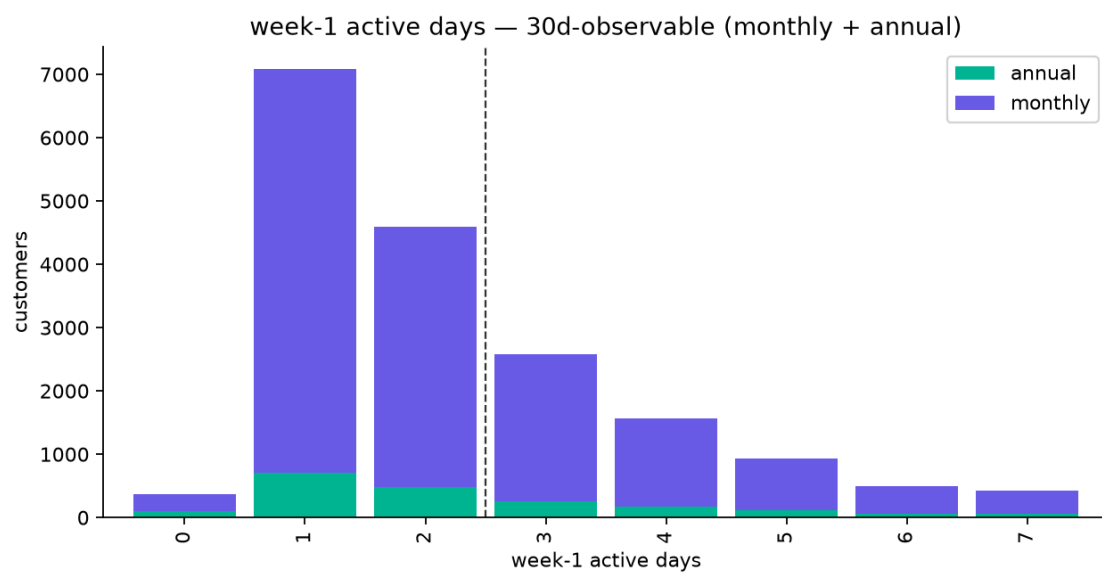
Base: **19,125** customers with tenure ≥ 32 d (monthly + annual), i.e. everyone whose first cycle has fully elapsed.

≤ 1 day in week 1 and nothing after	→ 1. one-shot	5,743 (30%)
< 3 days in week 1	→ 2. weak week-1	6,887 (36%)
≥ 3 days, jobs $< p80$ (88)	→ 3. habit, light	3,669 (19%)
≥ 3 days, $p80 \leq \text{jobs} \leq \text{tail}$	→ 4. habit, heavy	2,323 (12%)
≥ 3 days, top of jobs (Lorenz tail)	→ 5. top usage	503 (3%)



Tail = smallest top slice that holds 50% of jobs, floor p95 → **≥787 jobs** (top 3%). Annual is ~10% in every bucket — not a sixth segment.

Why 3 day 66% of people show up 0–2 days in week 1 (pile at 1). Expansion is dead at 0–1 (~7%), steps at day 2, then a slope — no cliff. ≥2 would call 59% “habit.” ≥4 looks stronger only because the leftover group is smaller, and it dumps 2.7k people (22% expand) into weak week-1. 3+ active days is the first observed behavioral threshold associated with materially higher subsequent engagement.



	1 one-shot	2 weak W1	3 habit light	4 habit heavy	5 top usage
ARPU_meq	\$22	\$31	\$34	\$71	\$138
credits ARPU	\$0	\$2	\$3	\$21	\$62
paid renewal	48%	51%	49%	55%	59%
used after W1 / M2	0 / 10%	62 / 38%	60 / 46%	89 / 73%	98 / 91%
w1 / m1 days	1 / 1	2 / 2	3 / 5	4 / 10	6 / 18
m1 jobs (after W1)	5 (0)	17 (3)	29 (2)	203 (75)	1,345 (864)

expand / up / credits	4 / 3 / 1%	14 / 9 / 7%	21 / 10 / 12%	45 / 24 / 31%	73 / 47 / 48%
jobs per \$	0.3	0.9	1.1	4.5	20.6

Renewal barely separates the segments — monetization does. Under the strict rule the whole base sits in a 48–59% band, so the best behaviour in the data buys ~11 pp of first renewal, not the 28 pp the loose window showed (46% → 74%). ARPU_meq over the same segments moves 6x (\$22 → \$138) and expansion 18x (4% → 73%). Behaviour is an *expansion* signal that only weakly predicts the auto-charge.

1–3 are not a value ladder. ARPU is \$22–34, renewal ~50% everywhere, jobs after week 1 are 0 / 3 / 2. They exist because of week-1 days: 1 never comes back, 2 showed up twice, 3 has a habit (3 days, then 5 in the month) but still looks like 2 on money and volume.

4 vs 5 is volume, then margin. Jobs 29 → 203 → 1,345. Expand 21 → 45 → 73%. Credits \$3 → \$21 → \$62. Jobs per \$ 1.1 → 4.5 → 20.6. Video share collapses (1.00 → 0.14 → 0.02) -> light users do a few video jobs; heavy users run image pipelines.

In 4–5, plan is the pay path, not the segment. Jobs are flat inside each bucket (190–230 in 4, ~1.3k in 5), so what changes is where the money can go. Entry-plan Basic converts on both routes (42% upgrade / 38% credits in 4; 83% / 58% in 5), Pro is in the middle (19% / 29%), Ultimate has no rung above it and mostly does neither (2% upgrade, 22% credits). Video share rises with the rung (0.04 Basic → 0.33 Ultimate in 4): the heaviest Basic accounts are image factories on the cheapest plan.

11. So what?

Structural differences

Across the 5 segments, there is a fundamental behavioral split:

- Activation split (1–3 vs 4–5). Segments 1–3 enter for video, generate little, and pay \$22–34 with ~50% passive renewal. Segments 4–5 build image-heavy production workflows, pay \$71–138 and drive nearly all expansion revenue. Renewal differs by only ~7 pp across the split — the gap is in money, not in survival.
- Within 1–3, the failure point moves: never returned (1) → returned but no routine (2) → routine but no monetization (3). One funnel, three stages — which is why the goals below are stage-specific but the segments share one North-Star: M2 active rate

$$ICE = I \times C / E$$

Segment 1: One-Shot

Share: 30.0% (n=5,743) | Paid Renewal: 48% | M1 Jobs: 5.0 (100% Video)

User profile

Users log in on Day 1, execute ~5 video jobs, their expectations are not met, understanding of the platform is low or , and leave permanently during Month 1. Despite a 48% passive renewal rate, usage in M2 drops to 10%.

Product Goal

Activation Recovery. Eliminate Day-1-3 video generation experience failure, establish a 1-week creation loop, and prevent passive auto-churn at M1 renewal.

Metrics Framework

Target Metric: share with 2 or more days containing a completed-and-downloaded generation in W1, sessions in W1

Guardrail Metric : Day-1 Video Cancellation Rate (< 5%).

Informational Metrics: first-job failure/cancel rate, time to first render, same-prompt re-rolls in session 1, CR generation → download

Hypothesis

#	Idea	Product / UI concept	Why it fits	Risks / costs	Test	I	C	E
1	Generation presets	Replace blank prompt box for new users with pre-configured viral presets, choice out of top popular	Fixes the effort needed to get expected output	Generic, not "theirs"; hides free-form; ignores intent	A/B	8	7	3
2	Prompt Enhancer	AI suggestion restructuring raw input into video-model syntax, shown inline	Removes the syntax skill needed for a usable clip + potential Upsell feature available to higher subscriptions	Rewrite loses intent; LLM latency + cost per gen	A/B	8	6	3
3	Fast-Render Mode	picture previews of some moments of the video before full generation	Buys iteration inside the one session, aligns with expectations	Extra compute; one more step	A/B; guardrail = D1 first-render conversion	7	5	5
4	Easier UI	special lending for new users with less features - to skip the mental load	Try first without thinking much	misses the feature user came for, less choice	A/B; guardrail = skip rate + D1 conversion	6	5	5

		of many options						
5	Onboarding tour	3-step tour on first login: idea → prompt → parameters	Forces success where product understanding is low	Skipped, annoying, delays the first gen	A/B; guardrail = skip rate + D1 conversion	5	4	4
6	Re-engagement with their own attempt	Day-3 email: their first clip + a high-performing prompt in the same style + 1 free retry	Personal artifact beats generic promo	Spam, unsubscribes; low engagement; free retry gifted to natural returners	A/B	3	3	2

Segment 2: Weak Week-1

Share: 36.0% (n=6,887) | Paid Renewal: 51% | M1 Jobs: 17.0 (100% Video)

Behavioral Diagnosis

Users return after Week 1, but usage remains sparse (~2 active days in W1, 17 total jobs in M1). Desktop usage is high (71%), but users lack a systematic workflow.

Product Goal

Habit Formation. Build a repeatable weekly creation routine by lowering the effort needed to start new projects.

Metrics Framework

Target Metric: Month 1 Active Days (m1_days from 2.0 to 5.0+).

Guardrail Metric: Paid Renewal Rate

Informational Metrics: gap days between sessions, share of sessions started from a draft/prior asset, session duration

Hypothesis

#	Idea	Product / UI concept	Why it fits weak week-1	Risks / costs	Test	I	C
1	Continue your project" widget	Home opens on their last clips and drafts with one-click "variation/improvement" instead of a blank box	solves the cold start problem; their own artifact is	Build on the main surface; useless if session 1 produced	A/B	8	6

			the re-entry hook	nothing worth continuing			
2	Image-to-Video	Create a base of images / character photos for users to use them as preset	Repeatable start with no new idea needed	Shifts volume to cheap image jobs — activity up, value flat	A/B	7	6
3	Project Workspace & Drafts UI	Desktop project manager saving unfinished renders and prompts	70% desktop, sparse sessions — resuming beats restarting	Build cost that pays only if they return; storage for dead drafts	Feature flag to 20% of segment; measure gap days	6	6
4	1-Click Camera Angle Tweaks	"Re-angle" button — same scene, new motion, no rewrite	Turns one idea into several jobs — the missing volume (17 in M1)	Re-rolls raise compute with no revenue	A/B	5	5
5	Desktop Push & Browser Notifications	desktop alerts when long video rendering jobs finish processing	Closes the loop on slow renders so a sparse session does not end at the wait	Notification fatigue and opt-outs; only helps people already in the tab; does nothing to later user activity	A/B	4	5
6	Weekly Content Trend Templates	Rotating weekly prompt packs for TikTok/Reels creators based on usage on platform/new releases	A scheduled reason to come back	Content treadmill; copycat output	A/B	4	4

Segment 3: Habit, Light

Share: 19.2% (n=3,669) | Paid Renewal: 49% | M1 Jobs: 29.0 (100% Video)

Behavioral Diagnosis

Highly consistent, healthy usage (3 days in W1, 5 days in M1). High contribution margin per user (jobs_per_usd = 1.07). However, 30-day renewal stagnates at 49% due to under-monetization or plan friction.

Product Goal

Retention & Monetization. Increase first-renewal retention and drive conversion to higher tiers/credit top-ups

Metrics Framework

Target Metric: M2 active rate

Guardrail Metric: contribution margin per user

Informational Metrics: Credit limit hit rate, Exploration of features/product (eg models and presets used)

Hypothesis

#	Idea	Product / UI concept	Why it fits habit, light	Risks / costs	Test	I	C	E	
1	Saved Custom Styles Library	Save own prompt templates and settings for reuse	Deepens the routine, raises switching cost	Slow payoff, no monetization on its own	Flag to 20%; measure M2 active	8	6	4	
2	Month-in-review before renewal	3 days pre-renewal: best clips, features	renews what user enjoyed about the platform	Same recap reminds churners to cancel	A/B	5	4	2	
3	Promotion	At M1 renewal: recap "you made X clips in Y sessions" + subscription promotion	Direct hit on the 49% renewal; a commitment by deleting future churn reasons	Discount to some who'd renew monthly anyway; cannibalization of users who'd have expanded	A/B	4	4	2	
4	Usage notifications + Credit Top-Up Promotion at 80% Quota	Contextual top up pack when credits left drop below 20%	Promotes expansion revenue while keeping the user in control	promotion never fires for 80–90% of the segment; small uplift	A/B	3	4	2	

Segment 4: Habit, Heavy

Share: 12.1% (n=2,323) | Paid Renewal: 55% | M1 Jobs: 203.0 (14% Video / 86% Image)

Behavioral Diagnosis

Commercial and heavy creators. 85% on Desktop. They expand heavily (\$20.86 in credits, \$114.31 ARPU_30d) and shift toward image-to-video prep and batch asset creation

Product Goal

Expansion & Professional Workflow Integration. Maximize Net Revenue Retention (NRR) by driving subscription tier upgrades (Pro \$→ Ultimate/Creator) and supporting professional multi-asset workflows

Metrics Framework

Target Metric : 30d Expansion ARPU, Upgrade Rate Guardrail Metric: Compute Cost per Active User / Gross Margin. Informational Metrics: Credit top-up frequency, Storyboard canvas usage, Character consistency feature usage

Hypothesis

#	Idea	Product / UI concept	Why it fits habit, heavy	Risks / costs	Test	I	C	E
1	Character & Asset Consistency	Lock character and style across image/video generations	83% image is asset prep before video passes — consistency is the pain in exactly that flow	Training/serving cost; likeness and moderation exposure	release on low num of users	8	8	6
2	Integrated Storyboard Canvas	Node/timeline desktop UI linking image gens into video tracks and segmenting them in projects	the pipeline already exists in behavior, helps users to organize	-	fake door test first	7	7	6
3	Credit bundle pricing	Bigger prepaid packs at better rates (\$50 / \$100), surfaced to repeat top-uppers	Zero-build monetization of a behavior that already exists; pulls revenue forward, raises commitment	Bulk discount cuts per-credit revenue if buyers were price-insensitive	A/B	3	5	2
4	Upgrade promotion	Modal only after 2	overpayers are the one	Mis-scoped it converts top-	A/B	3	4	2

		consecutive months of top-ups above the next-tier gap, Pro+ only (Basic excluded — their path is credits); shows the personal math	group where the upgrade is win-win: they save vs top-ups and generate committed revenue	up revenue into a cheaper flat plan — hence the 2-month rule					
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Segment 5: Active, Top Usage

Share: 2.6% (n=503) | Paid Renewal: 59% | M1 Jobs: 1,345.0 (2% Video / 98% Image)

Behavioral Diagnosis

Extreme power users operating at 20.64 jobs/USD (1,345 jobs/mo). High expansion (\$61.71 in credits, \$185.83 ARPU_30d). They run high-volume batch image processing and API-like generation flows.

Product Goal

Margin Guardrailing & Enterprise Conversion. Protect compute contribution margins by enforcing strict credit limits on flat plans while converting high-volume accounts to Custom Enterprise/Agency Tiers.

Metrics Framework

Target Metric : Contribution Margin per User & Credit/Overcharge Revenue

Guardrail Metric: churn, LTV.

Informational Metrics: cw-jobs per \$, enterprise contact requests, API waitlist signups

Hypothesis

#	Idea	Product / UI concept	Why it fits top usage	Risks / costs	Test	I	C	E	ICE
1	Self-serve "Max" tier above Ultimate	New top consumer plan: big credit allotment, higher concurrency, priority queue, early model access. Upsell banner	need already exists, it could be more profitable	Priced wrong it cannibalizes credit revenue (\$62 here)	Fake-door	8	7	3	18.7

2	Co-pilot workflow	In-canvas assistant that proposes and can run the next pipeline step (edit still → variants → video) as a batch, MCP connectors added	1,345 jobs/mo is already a production flow — the bottleneck is stitching steps, not finding an idea	LLM cost at this volume; over-automation can derail intent	pilot version	8	7	4	14.0
3	AI image references improvement	VLM feature, suggesting edits for photos attached as references to improve video generation quality	Most usage in this segment is in generating images for videos - improves experience in the main step	adds a step to user journey	pilot version	8	6	4	12
4	Automatic scenario generation	based on input images, generates a scenario based on context of previous chats	In huge volumes the main goal is to improve the outcome and speed	may change the user's intent, needs guardrails	pilot version	7	6	4	10.5
5	Model Efficiency Selector (Turbo vs HQ)	User-chosen fast/cheap vs high-compute models	Keeps volume while cost follows compute; explicit choice, not auto-routing — users may not trust auto	-	pilot version	6	6	4	9

No ideas in team plans/enterprise/etc, because this case if focused on B2C improvement

12. Overall prioritization

Ranked from the product bets in §11 — the feature that fixes each segment's diagnosed failure, then by how many customers it reaches. Marketing ideas (promos, recaps, packs) sit below: they leave the product unchanged. Where a dollar range is given it assumes the recent run-rate (~6.2k new paying accounts a month in Jul–Aug) and is order-of-magnitude.

Rank	Bet	Seg	Features	~impact	Effort	What to know first
1	First-generation experience	1	Generation presets, Prompt Enhancer	retention: activation on a 48% first renewal, 30% of the base	med	first-day interaction with output statistics and cancel rate
2	Max tier + Co-pilot workflow	5	Self-serve "Max" tier above Ultimate, Co-pilot workflow	expansion: turns part of the \$62/mo top-up spend into a plan; keeps a 1,345-job pipeline in-product → ~\$100–170k a year	med	whether Max can be priced above current top-up spend without cannibalising it
3	Continue your project	2	Continue your project widget, Image-to-Video	retention: habit — month-1 active days from 2 toward 5, 36% of the base	med	whether session 1 leaves anything worth continuing
4	Saved Custom Styles Library	3	Saved Custom Styles Library	retention: switching cost on a 49% renewal, M2 active rate	med	whether they already reuse prompts and settings
5	Production pipeline	4	Character & Asset Consistency, Integrated Storyboard Canvas	expansion: holds the 12% who already pay \$71 and expand 45%	high	cost per generation — breakeven here is \$0.12 per image job

Parked: Month-in-review before renewal, Promotion, Credit bundle pricing, Upgrade promotion, Weekly Content Trend Templates, Re-engagement with their own attempt, Desktop Push & Browser Notifications, Onboarding tour.

Why this order - by CJM and impact:

- First week: One-shot is 30% of the base and can contribute to LTV by retention growth if converted to active users
- Building habits on active users: Segment 5 has the other top product scores: Max converts visible willingness to pay, Co-pilot stitches the pipeline they already run. Segment 2 is the largest group — Continue your project and Image-to-Video are how sparse sessions become a routine.